Sentence Builder UI Design

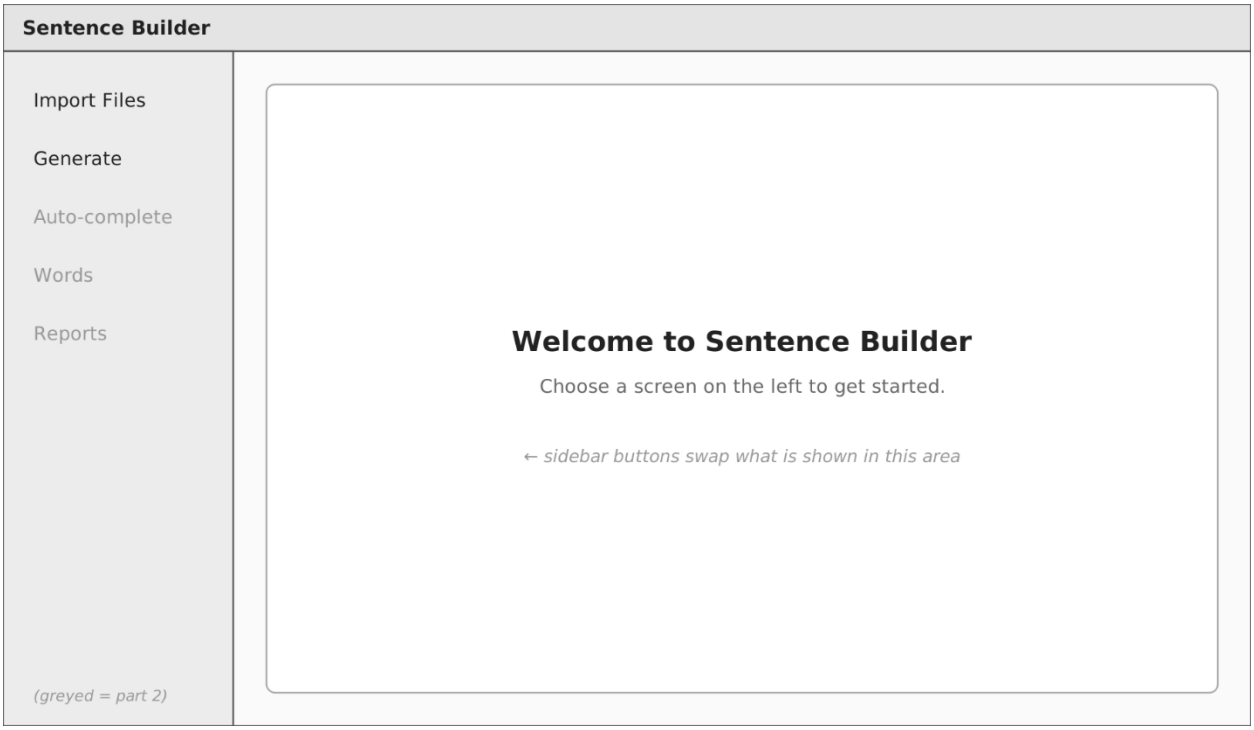
Overview:

This is part 1 of the UI design: the first 3 of the 6 screens (WBS 5.1–5.3), which are the ones needed for the Week 6 mid-project demo. Each screen uses only standard JavaFX controls (buttons, text fields, a dropdown, a progress bar, a table) so it can be built quickly in Scene Builder.

Screen 1

Main Window

One window with a sidebar of buttons on the left; clicking a button swaps the screen shown on the right. Nothing opens in a new window except the file picker.



(Image created in Claude)

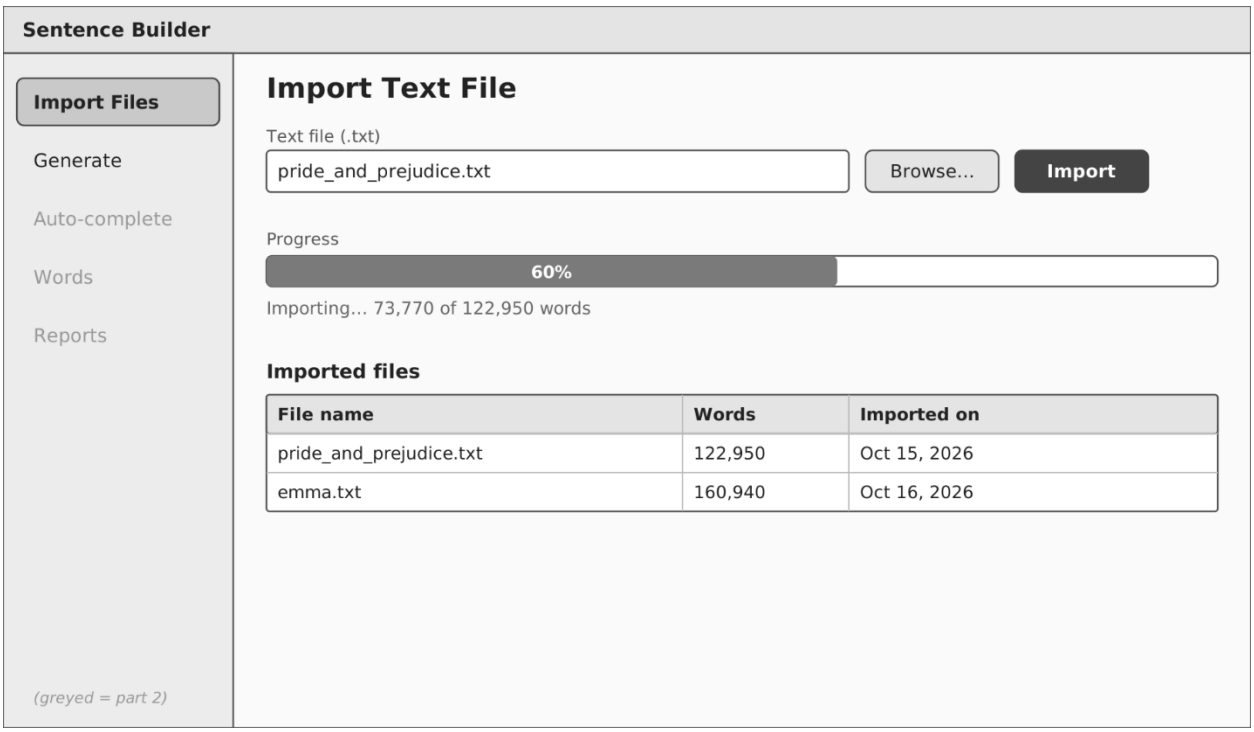
Part	JavaFX control
Window layout	BorderPane

Sidebar	VBox with 5 buttons
Screen area	Center of the BorderPane, swapped on button click
Welcome message	Two Lables

Screen 2

Import files

The user picks a .txt file, clicks Import, watches a progress bar, and sees the file added to a table of imported files.



(image created in claude)

Part	JavaFX control
File path	TextField (read-only)

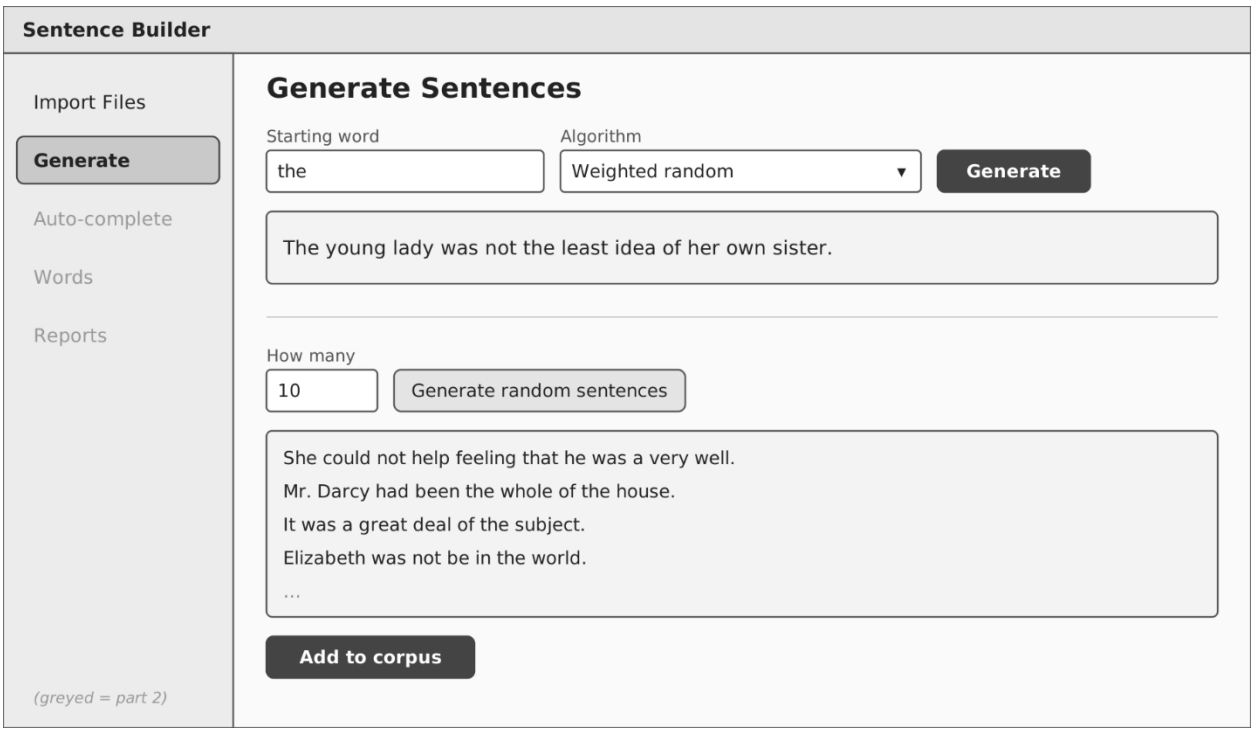
Browse	Button that opens a Filechooser limited to .txt
Import	Button
Progress	ProgressBar + Label
Imported files	TableView with 3 columns: file name, words, imported on

The import runs in the background, so the window doesn't freeze, and the progress bar updates as it goes.

Screen 3

Generate Sentences

The top half makes one sentence from a starting word; the bottom half makes several sentences from random words and adds them to the corpus with one button.



Part	JavaFX control
Starting word	TextField (read-only)
Algorithm	ComboBox listing the generator algorithms
Generate	Button
Result	Label
How many	TextField
Generate random sentences	Button
Sentence list	ListView
Add to corpus	Button

How it will be built

Each screen is one FXML file made in Scene Builder plus one controller class, so screens can be built and tested separately

File	Purpose
MainView.fxml + MainController	Window, sidebar, swapping screens
ImporttView.fxml+ImportController	Screen 2
GenerateView.fxml	Screen 3

Controllers only call the other modules (import, generator, database); they don't contain SQL or generation logic. Until those modules are ready, the screens use placeholder data.

Next Steps

1. Wireframes for screen 1 – 3
2. Team Review

3. Build the main window and sidebar in Scene Builder
4. Build the Import and Generate screens with placeholder data
5. Connect to the real import and generator modules
6. Part 2: wireframes for Auto-complete, Word Browser and Reports